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Autonomous Directional Drilling Revolutionizing Efficiency and Precision on Lumpsum Turnkey Rigs Through AI and Automation

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Abstract

Directional drilling plays a crucial role in wellbore trajectory control but traditionally requires continuous human intervention. On Lump Sum Turnkey (LSTK) rigs, where efficiency and precision are essential for meeting strict cost and schedule constraints, the adoption of digital technologies becomes imperative. This study investigates the implementation of an autonomous directional drilling system to enhance operational performance on LSTK rigs.

The autonomous system was deployed in Middle Eastern fields to automate the directional drilling process. Using advanced cloud-based software and intelligent systems, the technology establishes a continuous feedback loop between surface and downhole tools. It processes real-time trajectory data, evaluates multiple projection scenarios, determines the optimal trajectory, and transmits commands directly to downhole tools—eliminating the need for human intervention.

The system was evaluated on 29 BHA, drilling a total of 77,000 feet autonomously. Compared to offset wells, the system demonstrated a 24% increase in the rate of penetration, a 33% reduction in surface downlink commands, and a 22% reduction in well tortuosity. These improvements streamlined directional drilling and enhanced operational efficiency, facilitating smoother tripping and tubular running—key factors in LSTK projects.

This study highlights the potential of autonomous directional drilling to redefine turnkey operations by minimizing human dependency, enhancing precision, and improving overall efficiency. These findings reinforce the transformative role of automation and digitalization in optimizing wellbore trajectory management on LSTK rigs.

Summary for Conference Program

Autonomous directional drilling is revolutionizing efficiency and precision on LSTK rigs by reducing human intervention and leveraging intelligent systems. This study explores the deployment of an AI-driven system that automates the directional drilling process, enhancing performance through real-time trajectory analysis and automated downhole control. Field trials demonstrated significant efficiency gains, including a 24% increase in the rate of penetration and a 22% reduction in well tortuosity. By improving drilling accuracy and operational efficiency, this technology presents a scalable solution for the evolving demands

of LSTK projects, underscoring the transformative potential of AI and automation in modern drilling operations.

Introduction

Directional drilling is a specialized field within the broader practice of drilling deviated wells, serving a critical role in the efficient extraction of hydrocarbon reserves. Historically, this process has relied heavily on the expertise and judgment of directional drilling engineers to ensure precise trajectory control and alignment with the designated well plan (Gabriel et al., 2022; Oliveira et al., 2023). Engineers commonly employ steerable motor assemblies or rotary steerable assemblies to navigate wells along preplanned paths. The coordination of rotating and sliding sequences, or the transmission of downlink communication commands to rotary steerable tools, is meticulously managed by engineers to maintain trajectory accuracy and ensure compliance with the well plan.

A fundamental responsibility of directional drilling engineers is the continuous monitoring and updating of the well path. Well path projections provide real-time visibility into the actual position of the wellbore relative to the planned trajectory. Engineers use this information to make critical, timely decisions aimed at correcting deviations through sliding operations with motor assemblies or by issuing downlink commands for rotary steerable tools. However, variability in individual experience and decision-making practices can introduce inconsistencies in trajectory control, which in turn influence well path tortuosity and the time required to achieve optimal steering outcomes.

To address the challenges posed by human variability and its impact on efficiency, the industry is increasingly adopting digital solutions and advanced technologies to augment—and, in some cases, automate—directional drilling processes. The shift toward intelligent execution automation seeks to standardize and optimize operations, enhance trajectory precision, reduce the frequency of downlink commands, and minimize wellbore tortuosity. The adoption of digital technologies in directional drilling encompasses a wide range of innovations that drive improvements in operational efficiency and reliability. These technologies utilize real-time data analytics and cloud computing to automate decision-making processes that were previously dependent on human expertise. By incorporating intelligent algorithms into drilling workflows, digital solutions can process large volumes of data in real time, identify optimal steering strategies, and dynamically adjust operational parameters to maintain trajectory integrity.

This evolution toward intelligent execution automation is further supported by ongoing advancements in sensor technology and data integration capabilities. Modern sensors provide critical data inputs to automated systems, enabling precise control over drilling operations and supporting proactive decision-making to optimize wellbore placement. Figure 1 conceptually illustrates how the integration of digital technologies within the drilling domain can enable autonomous capabilities through computational systems that interface with and control the drilling process.

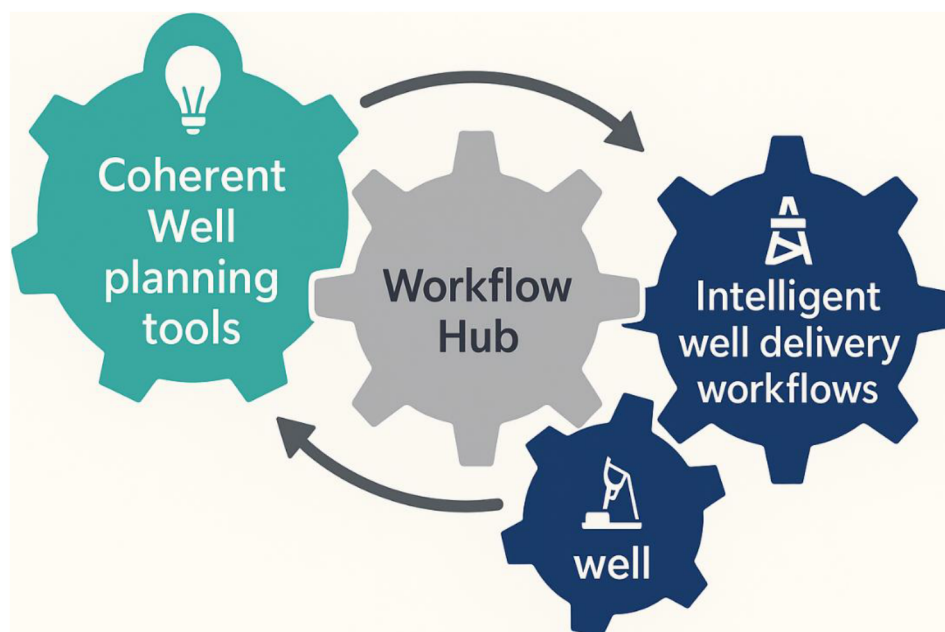


Figure 1—From planning to execution workflows to grow autonomous capabilities.

Maximizing the intelligent execution workflow for directional drilling steering requires automation capabilities that can seamlessly integrate with intelligent planning tools. This integration enables the consumption of well path data in a digital format, clearly defining both the planned trajectory requirements and operational tolerances. As a result, the full advantages of multisystem integration can be realized, delivering substantial efficiency improvements. Through this integration, continuous improvement initiatives can be effectively implemented throughout the drilling process.

This approach ensures that operational precision and performance are consistently optimized, in alignment with ongoing advancements in automated steering technologies and data-driven decision-making. Ultimately, the synergy between automated workflows and advanced planning tools enhances operational efficiency and fosters a proactive approach to managing directional drilling challenges. This alignment supports smoother execution of drilling strategies while establishing a foundation for continuous enhancements in drilling operations. Therefore, the integration of automation capabilities with intelligent planning tools is essential for achieving sustained improvements and operational excellence in directional drilling (Fig. 1).

Autonomous Directional Drilling Technology Overview

Autonomous directional drilling relies on several critical components to enable effective operation. These include a cloud-based directional drilling advisory application that hosts computational processes and a robust data engine. The advisory system communicates with an edge device that interfaces with the rig control system or other directional modules, allowing for automated workflow execution.

A key feature of the system is a dedicated driller interface, which enables drillers to review and authorize autonomous decisions. This interface also provides manual override capabilities, ensuring drillers can intervene when necessary. Together, these integrated components form a cohesive architecture supporting autonomous directional drilling.

The cloud-based application functions as the computational and data management hub, while the edge device ensures seamless connectivity and workflow execution. The driller interface maintains human oversight as a fundamental aspect of the system, offering real-time monitoring and control of automated processes to achieve optimal drilling outcomes drilling conditions and real-time challenges.

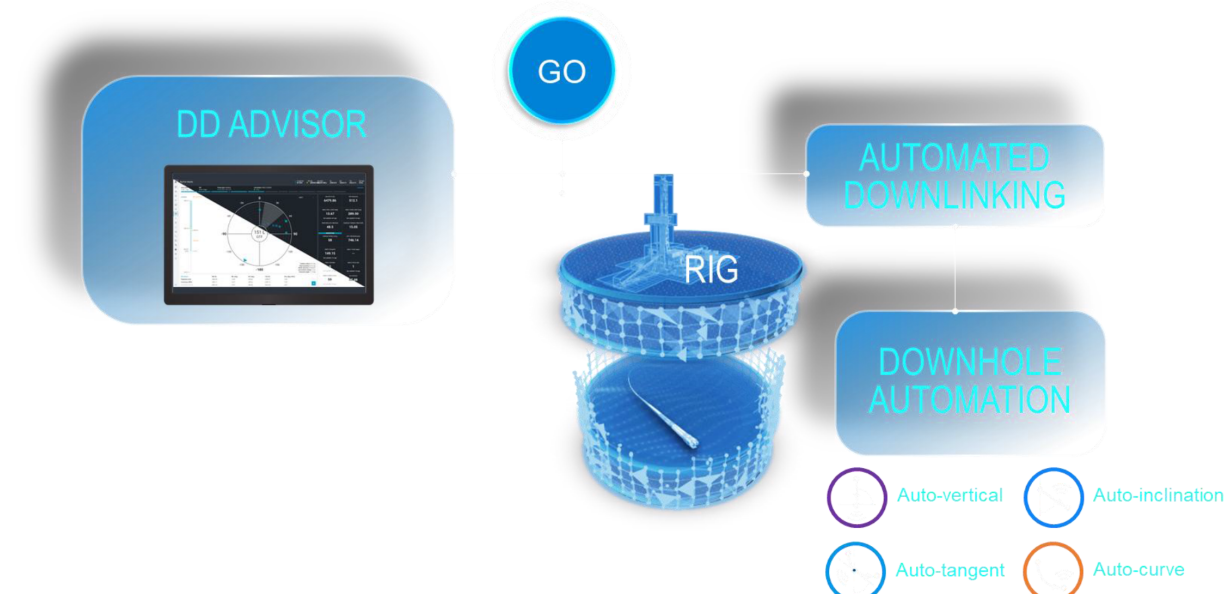


Figure 2—Autonomous directional drilling capabilities.

Autonomous Directional Drilling Steering Loop Workflow

The autonomous steering loop workflow represents a major advancement in directional drilling, utilizing advanced algorithms to automate and optimize steering of deviated wells. This process reduces reliance on directional drilling engineers by integrating real-time data processing and decision-making directly into the drilling workflow.

The system follows a structured workflow loop designed to ensure precise trajectory control and minimize wellbore tortuosity, thereby enhancing efficiency and accuracy. The workflow replicates the core tasks of a directional drilling engineer and consists of several key steps, as illustrated in Figure 4 and detailed below:

1. Communication to Surface

Downhole tools equipped with telemetry capabilities transmit essential directional data (e.g., inclination and azimuth) to the surface. This real-time communication allows the autonomous system to compute the wellbore's current position relative to the planned trajectory.

2. Position Assessment and Behavior Analysis

The transmitted data is used to compute the wellbore's exact subsurface position. The system compares the actual trajectory to the planned path, evaluates deviations, and identifies necessary corrections to maintain alignment with the target trajectory.

3. Path Optimization

The advisory system evaluates multiple potential well paths to determine an optimal trajectory that minimizes tortuosity while respecting operational and geological constraints. These computations are delivered via the cloud-based system, focusing on maintaining trajectory accuracy and drilling efficiency.

4. Command Planning and Optimization

Once an optimal path is established, the system plans and optimizes a sequence of steering commands to guide downhole tools along the trajectory. This includes determining directional tool yield, tool orientation adjustments, steering percentages, and other operational parameters required for precise control.

5. Communication to Downhole Tools

The optimized command sequence is transmitted to the rig's interface system or directional modules controlling the downhole tools. Rotary steerable systems receive directional instructions through adjustments in flow rate or revolutions per minute (RPM), with the cloud-based system directly interfacing with rig controls to ensure seamless and automatic execution.



Figure 3—Autonomous directional drilling steering loop.

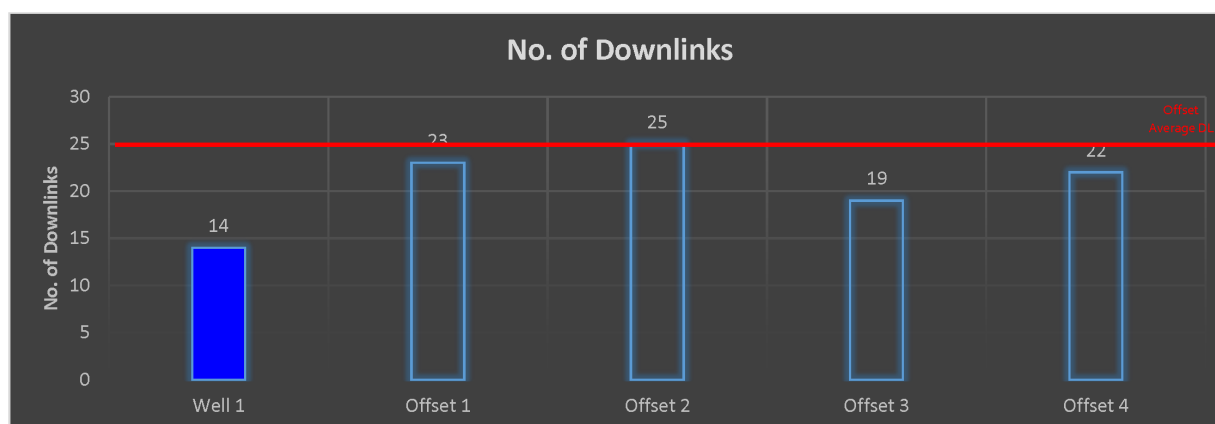


Figure 4—Autonomous system achieving downlinks reduction in Well 1.

Upon executing the commands, the system continuously monitors downhole tool responses and updates the wellbore location data in real time. If deviations or unexpected conditions occur, the system iteratively adjusts the trajectory plan and command sequence to maintain accuracy.

This structured workflow highlights the transformative impact of autonomous directional drilling in improving operational efficiency, reducing risks, and promoting consistency in drilling operations. By automating complex decision-making and leveraging advanced computational capabilities, the system minimizes human intervention, reduces outcome variability, and optimizes drilling performance.

Cloud Integration and Real-Time Optimization

Utilizing cloud computing, the system processes large volumes of data and performs computation-intensive tasks required for trajectory optimization in real time. It continuously analyzes the drilling plan and dynamically updates commands to maintain trajectory accuracy.

Integration with rigsite operations enables efficient command transmission, reducing latency and enhancing responsiveness. The cloud-based system communicates directly with the rig control system, ensuring seamless execution of steering commands and contributing to overall drilling performance.

This architecture exemplifies a shift toward automated, data-driven approaches in the oil and gas industry. Centralizing computation and command transmission not only optimizes drilling efficiency and accuracy but also supports operational safety, reduces environmental impact, and promotes resource optimization in drilling operations.

Field Implementation and Qualification

The autonomous directional drilling technology was endorsed by a major operator to assess its performance and quantify its value across different fields, well sections, and profiles. Between 2023 and 2025, the technology was deployed across 22 LSTK wells (see [Table 1](#)), covering a total of 77,000 feet. The initial deployment focused on rotary steerable system (RSS) runs.

Table 1—Summary of autonomous directional drilling wells delivered over 22 LSTK wells performance evaluation

Well Nu	Hole Size	Trajectory	Depth In	Depth Out	Footage	Autonomous footage	ROP	Field Avg ROP
1	6.125in	Lateral	9,382	15,435	6,053	5447.7	90	71.00
2	8.5in	Lateral	6,394	9,698	3,304	2478	55	55.00
3	8.375in	Curve	10,791	14,314	3,523	1409.2	39.23	25.70
4	8.375in	Curve	12,345	14,705	2,360	1888	35.71	25.70
5	16in	Curve	4,550	6,515	1,965	1473.75	31.7	31.00
6	16in	Curve	4,341	12,958	7,011	5258.25	30.78	31.00
7	8.375in	Curve	10,015	12,093	2,078	2078	30.56	25.70
8	8.375in	Curve	9,343	11,099	1,756	1756	28	25.70
9	12in	S-Shape	10,933	12,796	1,863	745.2	25.92	23.00
10	12in	Vertical	6,705	10,756	4,051	3807.94	35.33	21.49
11	16in	Curve	4,550	6,420	1,870	1402.5	25	31.00
12	8.375in	Curve	11,008	11,385	377	377	23	25.70
13	16in	Curve	4,360	10,387	6,027	5725.65	22.64	32.29
14	16in	Curve	6,017	10,432	4,415	3399.55	22.38	32.29
15	16in	Curve	4,637	11,121	6,484	5835.6	21.87	32.29
16	5.875in	Lateral	14,315	20,300	5,985	4369.05	18.94	20.38
17	5.875in	Lateral	14,705	18,242	3,537	1768.5	18.86	14.95
18	5.875in	Lateral	12,686	18240	5,554	4165.5	17	17.96
19	5.875in	Curve	11,096	11,968	872	872	15	20.38
20	5.875in	Vertical	12,960	15,439	2,479	2479	15	14.64
21	5.875in	Curve	10,767	10,885	118	94.4	13.54	20.38
24	5.875in	Curve	15,665	16,966	1,301	1170.9	9.4	23.57

As part of the qualification process, three sections were evaluated in detail to measure improvements in drilling performance, reduction in downlink frequency, and reduction in wellbore tortuosity. The goal was to validate that the autonomous system could autonomously deliver various directional profiles to target specifications while achieving operational improvements and efficiency gains.

Case Study 1—Well 1

In this evaluation, the target was a 6 1/8-in Lateral section using RSS, LWD BHA through Geosteering application, drilled footage 5447.7 ft with 90% autonomous ratio, with flow downlink control

Results included:

- Perfect match between steering performance and required commands
- 40% reduction in downlinks compared to offset wells (Fig. 4)
- 26.76% improvement in ROP compared to offset wells (Fig. 5)

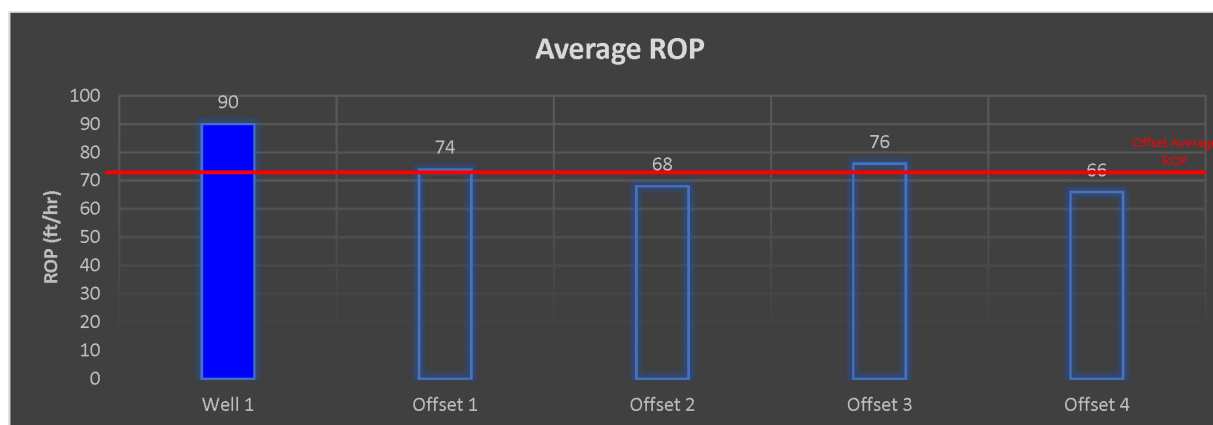


Figure 5—Autonomous system achieving better ROP in Well 1.

Case Study 2—Well 3

The second evaluation involved a 8 3/8-in. curve section with RSS, MWD BHA, drilled footage 3523 ft autonomously in a single run, achieving the planned trajectory with flow downlink control.

Key results:

- Steering performance fully matched the required commands (Fig. 6)
- 28% reduction in the total number of downlinks (Fig. 7)
- 52.65% improvement in ROP compared to offset wells (Fig. 8)

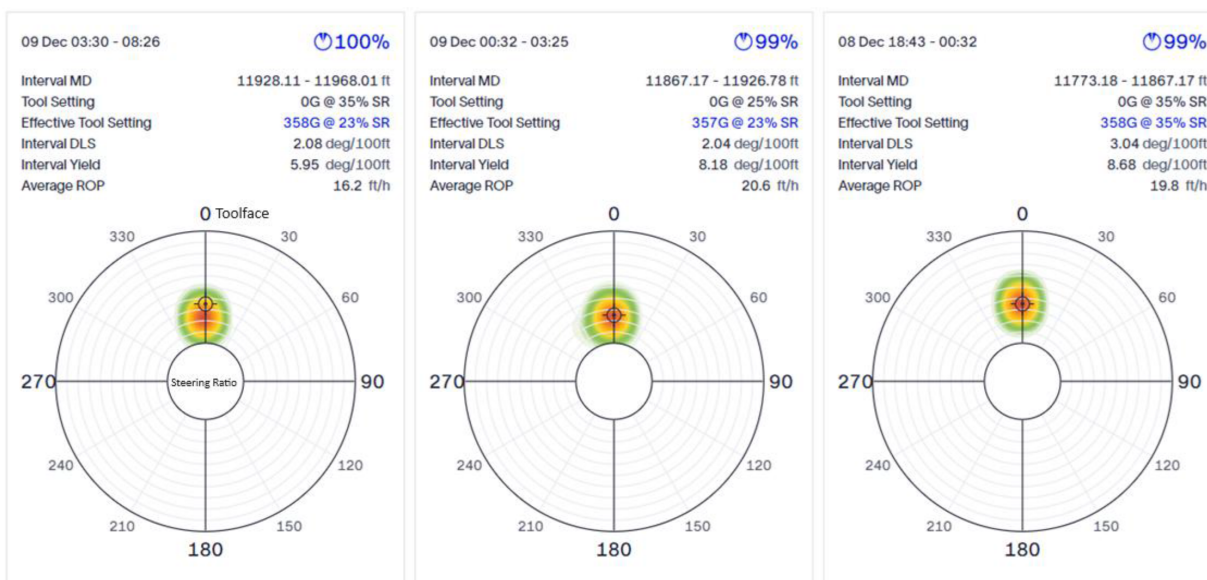


Figure 6—Steering performance matching with required commands in Well 3. The effective steering zone matches with target steering point.

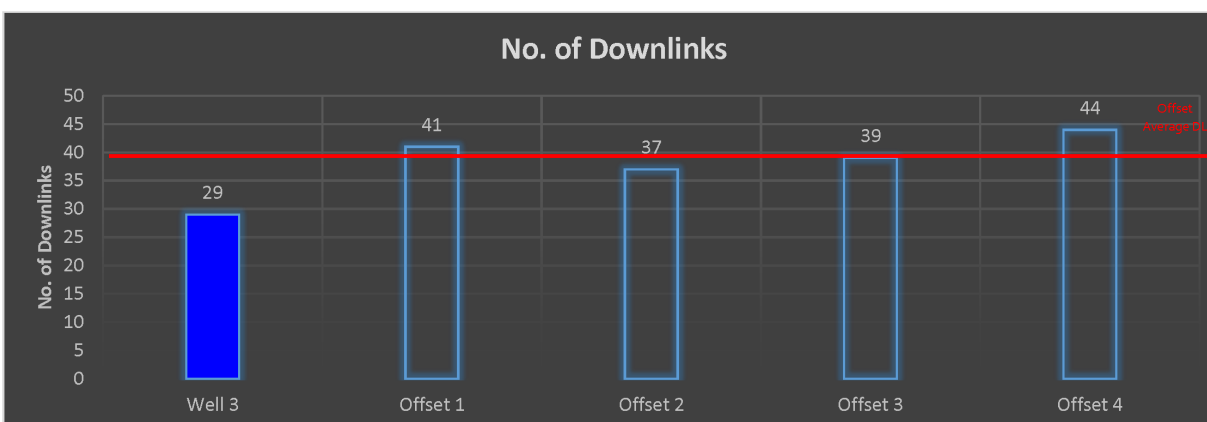


Figure 7—Autonomous system achieving downlinks reduction in Well 3.

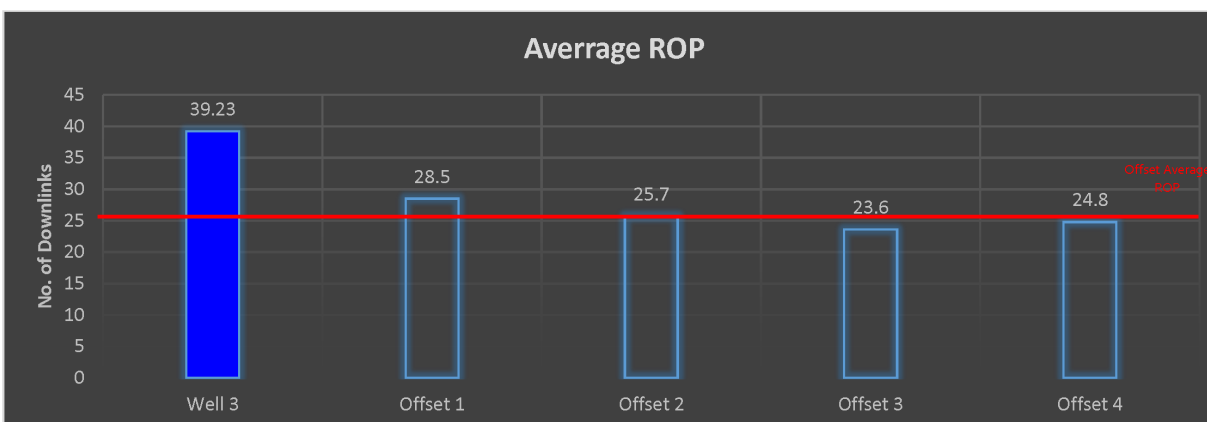


Figure 8—Autonomous system achieving better ROP in Well 1.

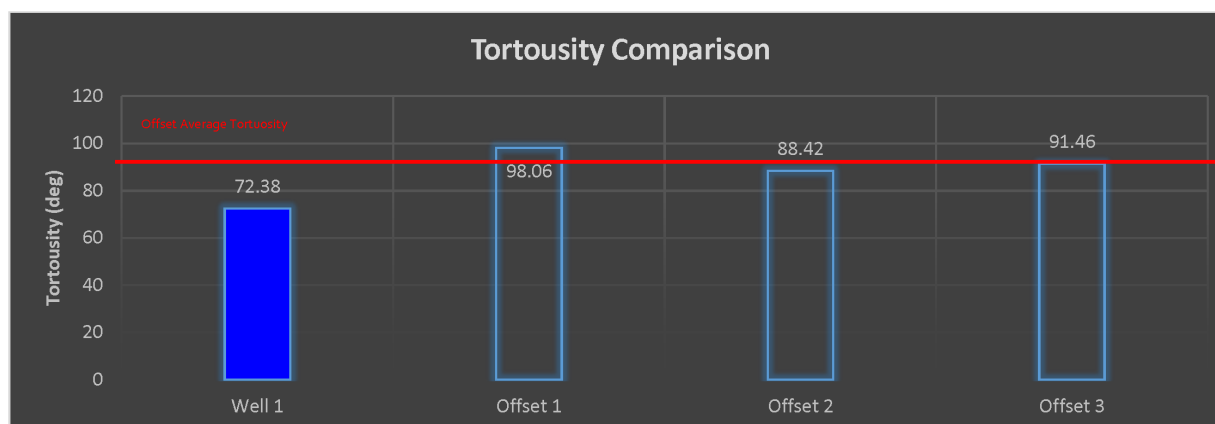


Figure 9—Autonomous system achieving tortuosity reduction in Well 1.

Technology Proven Benefits and Applications

The autonomous directional drilling system represents a significant technological advancement in the oil and gas industry. It integrates sophisticated command advisory and downlink control features that enhance drilling operations. Developed specifically to reduce downlink frequency and minimize wellbore tortuosity, this technology ultimately improves overall drilling performance while addressing key operational challenges such as torque and drag, bottomhole assembly (BHA) integrity, casing trippability, and the risk of mechanical sticking.

Through a series of rigorous evaluation runs, the system demonstrated measurable benefits:

- ~40% reduction in downlink frequency
- ~28% reduction in wellbore tortuosity

Specific field results highlighted:

- Well 1: 21.8% reduction in tortuosity (Fig. 09)
- Well 3: 13% reduction in tortuosity (Fig. 10)

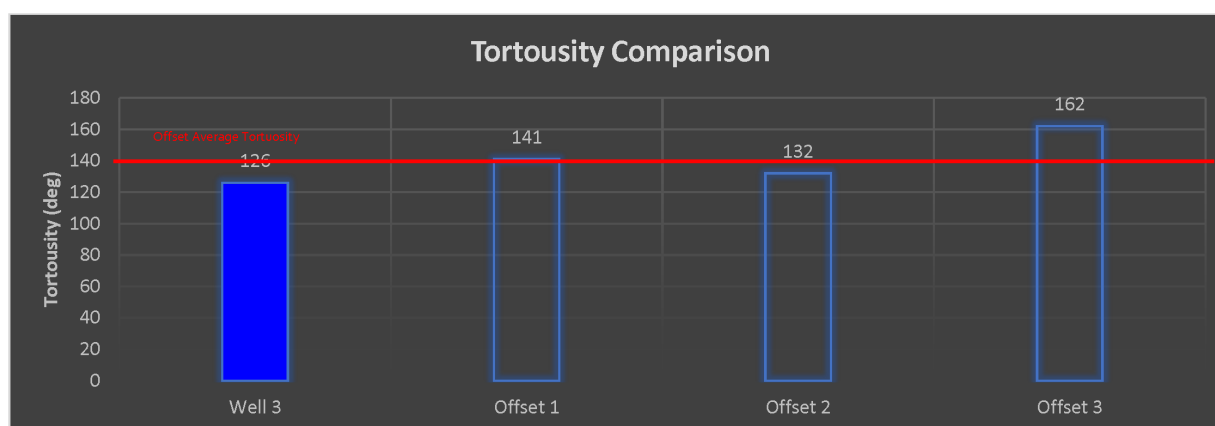


Figure 10—Autonomous system achieving tortuosity reduction in Well 3.

By improving trajectory control and reducing tortuosity, the technology enhances drilling accuracy and operational efficiency while minimizing time and resources traditionally spent on manual downlink communications. The result is consistently higher well quality with reduced tortuosity.

Economic and Environmental Benefits

In addition to improving technical performance, the implementation of autonomous directional drilling has demonstrated clear economic advantages and environmental gains:

The system generated an average of 1 rig day saved per well for the major operator.

It contributed to a 25 metric ton reduction in CO₂ emissions per well, thanks to improved drilling efficiency and reduced invisible lost time.

By optimizing resource utilization, the system supports both operational excellence and environmental sustainability within the energy sector.

Safety and Operational Efficiency

Beyond economic and environmental impact, the adoption of autonomous directional drilling enhances operational safety and efficiency:

Remote operation reduces the on-site crew footprint, improving safety by minimizing personnel exposure to hazardous rig environments.

The technology enables a single directional drilling engineer to remotely monitor and manage operations across multiple rigs simultaneously, illustrating the potential for streamlined operations and substantial manpower efficiency gains.

Driving Industry Transformation

The shift toward autonomous directional drilling aligns with broader Industry 4.0 principles, where digital technologies and automation reshape operational standards. By reducing human error and operational risk, autonomous systems improve drilling performance and support a safer, more sustainable approach to hydrocarbon extraction.

These advancements position autonomous directional drilling as a key enabler of the industry's digital transformation, promoting safer operations, greater efficiency, and improved environmental stewardship across global energy markets.

Conclusion

The results of this study clearly demonstrate that autonomous directional drilling technology is no longer conceptual—it is now a proven reality, delivering substantial benefits across the full spectrum of directional drilling operations.

Field trials have shown that autonomous directional drilling offers a consistent, system-based approach to optimizing directional control. By removing variability introduced by human behavior, the technology helps:

- Reduce wellbore tortuosity
- Improve operational efficiency
- Enhance drilling performance

Technical Achievements

From a technical standpoint, the system leverages advanced algorithms that analyze real-time data from downhole sensors and surface measurements. This data-driven approach enables the continuous optimization of directional drilling parameters, resulting in:

- Precise wellbore placement

- Minimized deviations from planned trajectories

The system successfully demonstrated Level 4 automation in directional drilling, where surface and downhole systems integrate seamlessly to optimize well path execution (Fig. 11). This level of automation is feasible and field-proven.

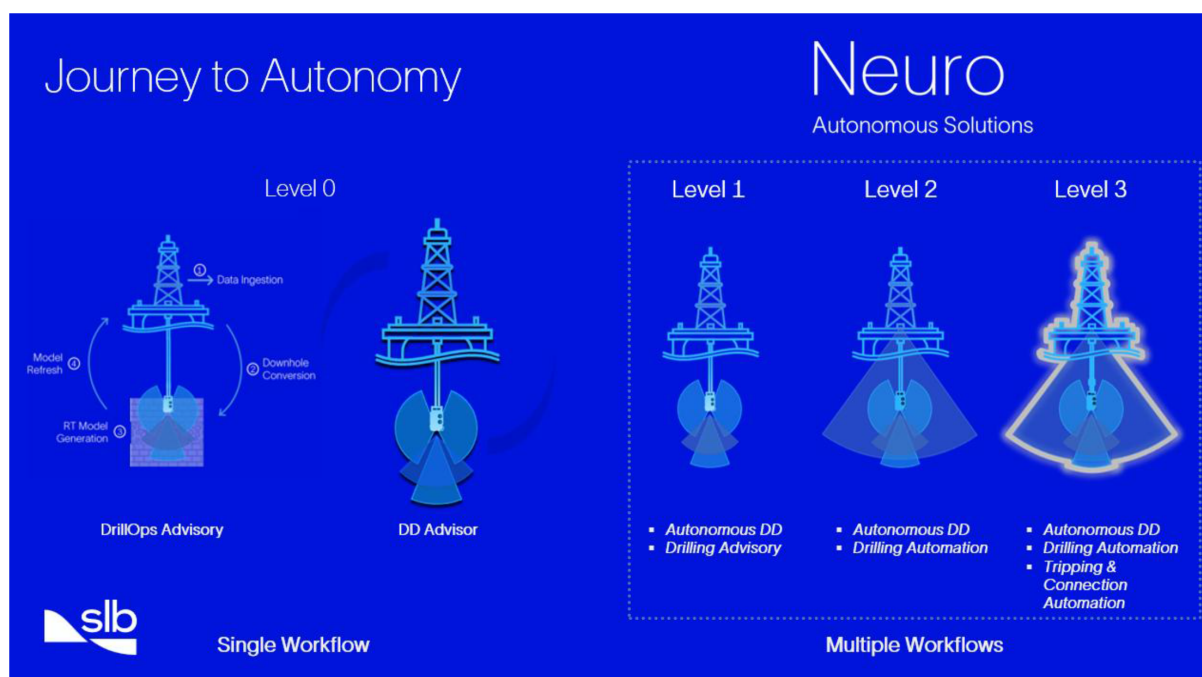


Figure 11—Autonomous directional drilling archives Level 4 autonomy through integration of surface and downhole systems.

Provides measurable improvements in drilling outcomes

The strategic value of Autonomous directional drilling represents a pivotal advancement for the energy sector. Its benefits span operational, environmental, and safety domains:

- Operational: Improved efficiency, optimized well paths, and reduced reliance on manual intervention.
- Environmental: Reduced CO₂ emissions through optimized resource use and efficient drilling practices.
- Safety: Enhanced operational safety by enabling remote monitoring and reducing the number of rigsite personnel required.

And the Future Outlook As the industry continues its digital transformation, autonomous drilling systems will play a central role in shaping the future of energy exploration and production. These technologies:

- Support broader goals of sustainability and environmental stewardship by reducing the industry's carbon footprint.
- Help drive continuous innovation by enabling data-driven decision-making and adaptive workflows.

By optimizing drilling paths and reducing unnecessary movements, the system conserves resources and minimizes energy consumption. The demonstrated reduction in CO₂ emissions further reinforces its value in supporting global climate goals and advancing sustainable energy practices.

In conclusion, autonomous directional drilling is a transformative enabler for the oil and gas industry, delivering superior performance, improved sustainability, and enhanced safety. Its scalability and adaptability will continue to drive innovation across the global energy landscape.

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